

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

Assessment Tool Weightage: 10%

. Detection Performance Comparison

Question:

Two object detection models produce the following results:

Model A (Viola-Jones): True Positives = 80, False Positives = 20, False Negatives = 30

Model B (YOLO): True Positives = 90, False Positives = 15, False Negatives = 20

(a) Compute precision and recall for both models.

(b) Compare their performance and state which model is better for object detection.

Answer:

(a) Precision and Recall

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

For Model A:

$$\text{Precision} = 80 / (80 + 20)$$

$$\text{Precision} = 80 / 100$$

$$\text{Precision} = 0.80 = 80\%$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

$$\text{Recall} = 80 / (80 + 30)$$

$$\text{Recall} = 80 / 110$$

$$\text{Recall} = 0.7273 = 72.73\%$$

For Model B:

$$\text{Precision} = 90 / (90 + 15)$$

$$\text{Precision} = 90 / 105$$

$$\text{Precision} = 0.8571 = 85.71\%$$

$$\text{Recall} = 90 / (90 + 20)$$

$$\text{Recall} = 90 / 110$$

$$\text{Recall} = 0.8182 = 81.82\%$$

(b) Comparison and Evaluation

Model A has a precision of 80% and recall of 72.73%.

Model B has a precision of 85.71% and recall of 81.82%.

Model B performs better because it has higher precision and higher recall.

Conclusion: Model B (YOLO) is better for object detection because it produces fewer false positives and detects more actual objects.

2. Motion Estimation Error Analysis

Question:

Two motion estimation techniques produce displacement errors (in pixels):

Frame 1: Method A = 2, Method B = 3

Frame 2: Method A = 3, Method B = 5

Frame 3: Method A = 2, Method B = 4

(a) Compute average error for both methods.

(b) Compare and evaluate which method is more accurate.

Answer:

(a) Average Error

For Method A:

$$\text{Average Error} = (2 + 3 + 2) / 3$$

$$\text{Average Error} = 7 / 3$$

$$\text{Average Error} = 2.33 \text{ pixels}$$

For Method B:

$$\text{Average Error} = (3 + 5 + 4) / 3$$

$$\text{Average Error} = 12 / 3$$

$$\text{Average Error} = 4 \text{ pixels}$$

(b) Comparison and Evaluation

Method A has an average error of 2.33 pixels.

Method B has an average error of 4 pixels.

Lower error means better accuracy.

Conclusion: Method A is more accurate because its average error is lower.

3. Optical Flow Magnitude Comparison

Question:

Optical flow magnitudes (in pixels/frame):

Region R1: Method A = 5, Method B = 8

Region R2: Method A = 6, Method B = 10

Region R3: Method A = 5, Method B = 9

(a) Compute average motion magnitude for both methods.

(b) Compare and evaluate which method captures stronger motion.

Answer:

(a) Average Motion Magnitude

For Method A:

$$\text{Average} = (5 + 6 + 5) / 3$$

$$\text{Average} = 16 / 3$$

$$\text{Average} = 5.33 \text{ pixels/frame}$$

For Method B:

$$\text{Average} = (8 + 10 + 9) / 3$$

$$\text{Average} = 27 / 3$$

$$\text{Average} = 9 \text{ pixels/frame}$$

(b) Comparison and Evaluation

Method A has an average motion magnitude of 5.33 pixels/frame.

Method B has an average motion magnitude of 9 pixels/frame.

Conclusion: Method B captures stronger motion because it has the higher average motion magnitude.

4. Depth Estimation Comparison (Stereo vs SfM)

Question:

Estimated depth values (in meters):

Object A: Stereo Vision = 2.0, Structure from Motion = 2.5

Object B: Stereo Vision = 3.0, Structure from Motion = 3.8

Object C: Stereo Vision = 4.0, Structure from Motion = 5.0

Actual depths: A = 2, B = 3, C = 4

(a) Compute absolute error for both methods.

(b) Compare and evaluate which method is more accurate.

Answer:

(a) Absolute Error

Absolute Error = |Estimated Value - Actual Value|

Stereo Vision:

Object A = $|2.0 - 2| = 0$ m

Object B = $|3.0 - 3| = 0$ m

Object C = $|4.0 - 4| = 0$ m

Average Stereo Error = $(0 + 0 + 0) / 3$

Average Stereo Error = 0 m

Structure from Motion:

$$\text{Object A} = |2.5 - 2| = 0.5 \text{ m}$$

$$\text{Object B} = |3.8 - 3| = 0.8 \text{ m}$$

$$\text{Object C} = |5.0 - 4| = 1.0 \text{ m}$$

$$\text{Average SfM Error} = (0.5 + 0.8 + 1.0) / 3$$

$$\text{Average SfM Error} = 0.77 \text{ m}$$

(b) Comparison and Evaluation

Stereo Vision has an average error of 0 m.

Structure from Motion has an average error of 0.77 m.

Conclusion: Stereo Vision is more accurate because it has zero error for all three objects.

5. Pattern Recognition Accuracy Comparison

Question:

Two recognition systems produce results:

Traditional: Correct Predictions = 85, Total Samples = 100

Deep Learning: Correct Predictions = 92, Total Samples = 100

(a) Compute accuracy for both systems.

(b) Compare and evaluate which system performs better.

Answer:

(a) Accuracy

Accuracy = Correct Predictions / Total Samples $\times$ 100

Traditional:

Accuracy = $85 / 100 \times 100$

Accuracy = 85%

Deep Learning:

$$\text{Accuracy} = 92 / 100 \times 100$$

$$\text{Accuracy} = 92\%$$

(b) Comparison and Evaluation

Traditional system accuracy = 85%.

Deep Learning accuracy = 92%.

Conclusion: Deep Learning performs better because it has 92% accuracy compared with 85% for the Traditional system.

6. Precision-Recall Trade-off Analysis

Question:

Two models produce:

Model A: TP = 70, FP = 30, FN = 20

Model B: TP = 85, FP = 25, FN = 25

(a) Compute precision and recall for both models.

(b) Compare and evaluate which model is more balanced.

Answer:

(a) Precision and Recall

For Model A:

$$\text{Precision} = 70 / (70 + 30)$$

$$\text{Precision} = 70\%$$

$$\text{Recall} = 70 / (70 + 20)$$

$$\text{Recall} = 70 / 90$$

$$\text{Recall} = 77.78\%$$

For Model B:

$$\text{Precision} = 85 / (85 + 25)$$

$$\text{Precision} = 85 / 110$$

$$\text{Precision} = 77.27\%$$

$$\text{Recall} = 85 / (85 + 25)$$

$$\text{Recall} = 85 / 110$$

$$\text{Recall} = 77.27\%$$

(b) Comparison and Evaluation

Model A has 70% precision and 77.78% recall.

Model B has 77.27% precision and 77.27% recall.

Conclusion: Model B is more balanced because its precision and recall are equal and it has higher precision than Model A.

7. F1-Score Comparison

Question:

Two models:

Model A: Precision = 0.8, Recall = 0.6

Model B: Precision = 0.75, Recall = 0.75

(a) Compute F1-score for both models.

(b) Compare and evaluate which model is better.

Answer:

(a) F1-Score

$$F1 = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$$

For Model A:

$$F1 = 2 \times 0.8 \times 0.6 / (0.8 + 0.6)$$

$$F1 = 0.96 / 1.4$$

$$F1 = 0.686$$

$$F1 = 68.57\%$$

For Model B:

$$F1 = 2 \times 0.75 \times 0.75 / (0.75 + 0.75)$$

$$F1 = 1.125 / 1.5$$

$$F1 = 0.75$$

$$F1 = 75\%$$

(b) Comparison and Evaluation

Model A has an F1-score of 68.57%.

Model B has an F1-score of 75%.

Conclusion: Model B is better because it has the higher F1-score and provides a better balance between precision and recall.

8. Optical Flow Consistency

Question:

Motion values:

Frame 1: Method A = 4, Method B = 6

Frame 2: Method A = 5, Method B = 8

Frame 3: Method A = 6, Method B = 9

(a) Compute average motion.

(b) Compare consistency and evaluate which is more stable.

Answer:

(a) Average Motion

For Method A:

$$\text{Average} = (4 + 5 + 6) / 3$$

$$\text{Average} = 15 / 3$$

$$\text{Average} = 5$$

For Method B:

$$\text{Average} = (6 + 8 + 9) / 3$$

$$\text{Average} = 23 / 3$$

$$\text{Average} = 7.67$$

(b) Comparison and Evaluation

Method A has a range of:

$$6 - 4 = 2$$

Method B has a range of:

$$9 - 6 = 3$$

Method A has less variation than Method B.

Conclusion: Method A is more stable because its motion values show less variation across frames.

9. Detection Accuracy Comparison

Question:

Results:

Model A: Correct = 180, Total = 200

Model B: Correct = 170, Total = 200

(a) Compute accuracy.

(b) Compare and evaluate.

Answer:

(a) Accuracy

For Model A:

$$\text{Accuracy} = 180 / 200 \times 100$$

$$\text{Accuracy} = 90\%$$

For Model B:

$$\text{Accuracy} = 170 / 200 \times 100$$

$$\text{Accuracy} = 85\%$$

(b) Comparison and Evaluation

Model A has 90% accuracy.

Model B has 85% accuracy.

Conclusion: Model A performs better because its accuracy is 90%, which is higher than Model B's 85%.

10. Error Rate Comparison

Question:

Results:

Method A: Errors = 20, Total = 100

Method B: Errors = 10, Total = 100

(a) Compute error rate.

(b) Compare and evaluate.

Answer:

(a) Error Rate

$$\text{Error Rate} = \text{Errors} / \text{Total} \times 100$$

For Method A:

$$\text{Error Rate} = 20 / 100 \times 100$$

$$\text{Error Rate} = 20\%$$

For Method B:

$$\text{Error Rate} = 10 / 100 \times 100$$

$$\text{Error Rate} = 10\%$$

(b) Comparison and Evaluation

Method A has a 20% error rate.

Method B has a 10% error rate.

Conclusion: Method B is better because it has a lower error rate.

11. Motion Estimation Variance

Question:

Errors:

Frame 1: $A = 2$, $B = 6$

Frame 2: $A = 3$, $B = 7$

Frame 3: $A = 2$, $B = 5$

(a) Compute mean error.

(b) Compare variability and evaluate stability.

Answer:

(a) Mean Error

For Method A:

$$\text{Mean} = (2 + 3 + 2) / 3$$

$$\text{Mean} = 7 / 3$$

$$\text{Mean} = 2.33$$

For Method B:

$$\text{Mean} = (6 + 7 + 5) / 3$$

$$\text{Mean} = 18 / 3$$

$$\text{Mean} = 6$$

(b) Variability

For Method A:

$$\text{Variance} = 0.22 \text{ approximately}$$

For Method B:

Variance = 0.67 approximately

Method A has lower variability than Method B.

Conclusion: Method A is more stable because it has lower mean error and lower variance.

12. IOU Comparison

Question:

IOU values:

Object 1: A = 0.6, B = 0.8

Object 2: A = 0.7, B = 0.85

Object 3: A = 0.65, B = 0.9

(a) Compute average IOU.

(b) Compare and evaluate detection quality.

Answer:

(a) Average IOU

For Method A:

$$\text{Average IOU} = (0.6 + 0.7 + 0.65) / 3$$

$$\text{Average IOU} = 1.95 / 3$$

$$\text{Average IOU} = 0.65$$

For Method B:

$$\text{Average IOU} = (0.8 + 0.85 + 0.9) / 3$$

$$\text{Average IOU} = 2.55 / 3$$

$$\text{Average IOU} = 0.85$$

(b) Comparison and Evaluation

Method A has an average IOU of 0.65.

Method B has an average IOU of 0.85.

Conclusion: Method B provides better detection quality because it has the higher average IOU.

13. Recall Comparison in Detection

Question:

Results:

Model A: TP = 60, FN = 40

Model B: TP = 80, FN = 20

(a) Compute recall.

(b) Compare and evaluate sensitivity.

Answer:

(a) Recall

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

For Model A:

$$\text{Recall} = 60 / (60 + 40)$$

$$\text{Recall} = 60 / 100$$

$$\text{Recall} = 60\%$$

For Model B:

$$\text{Recall} = 80 / (80 + 20)$$

$$\text{Recall} = 80 / 100$$

$$\text{Recall} = 80\%$$

(b) Comparison and Evaluation

Model A has 60% recall.

Model B has 80% recall.

Conclusion: Model B has better sensitivity because it detects 80% of the actual objects compared with 60% for Model A.

14. Depth Estimation Error (Squared)

Question:

Predicted values:

Object 1: Method A = 2.5, Method B = 2.2, Actual = 2

Object 2: Method A = 3.5, Method B = 3.1, Actual = 3

Object 3: Method A = 4.5, Method B = 4.2, Actual = 4

(a) Compute squared error.

(b) Compare and evaluate accuracy.

Answer:

(a) Squared Error

Squared Error = (Predicted Value - Actual Value)²

Object 1:

Method A = $(2.5 - 2)^2 = 0.25$

Method B = $(2.2 - 2)^2 = 0.04$

Object 2:

Method A = $(3.5 - 3)^2 = 0.25$

Method B = $(3.1 - 3)^2 = 0.01$

Object 3:

Method A = $(4.5 - 4)^2 = 0.25$

$$\text{Method B} = (4.2 - 4)^2 = 0.04$$

Average squared error for Method A:

$$\text{Average} = (0.25 + 0.25 + 0.25) / 3$$

$$\text{Average} = 0.25$$

Average squared error for Method B:

$$\text{Average} = (0.04 + 0.01 + 0.04) / 3$$

$$\text{Average} = 0.03$$

(b) Comparison and Evaluation

Method A has an average squared error of 0.25.

Method B has an average squared error of 0.03.

Lower squared error indicates better accuracy.

Conclusion: Method B is more accurate because its average squared error is lower.

15. Recognition Precision Comparison

Question:

Results:

System A: TP = 90, FP = 10

System B: TP = 85, FP = 5

(a) Compute precision.

(b) Compare and evaluate performance.

Answer:

(a) Precision

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

For System A:

$$\text{Precision} = 90 / (90 + 10)$$

$$\text{Precision} = 90 / 100$$

$$\text{Precision} = 90\%$$

For System B:

$$\text{Precision} = 85 / (85 + 5)$$

$$\text{Precision} = 85 / 90$$

$$\text{Precision} = 0.9444$$

$$\text{Precision} = 94.44\%$$

(b) Comparison and Evaluation

System A has a precision of 90%.

System B has a precision of 94.44%.

Conclusion: System B performs better because it has higher precision. Although System B has fewer true positives, it produces fewer false positives and therefore achieves a precision of 94.44%.

. Code of Conduct:

I K.L.V.S KARTHIK (192424177) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K.L.V.S KARTHIK

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant

Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation ; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand